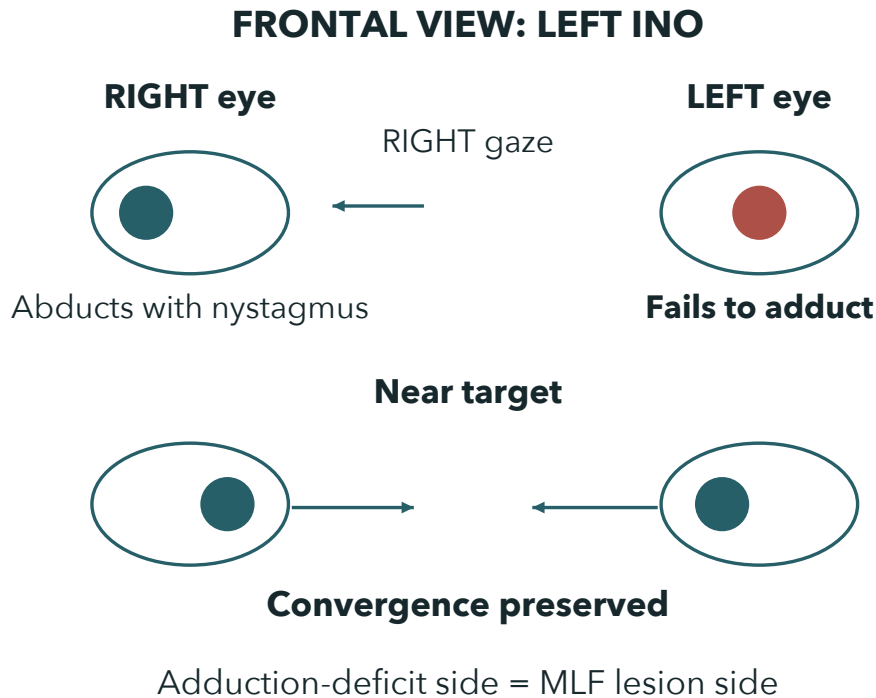


Diplopia: nerve patterns versus restriction

CLINICAL TYPE B | Focused decisions for M3-M5

A gaze pattern localises the problem

Conjugate gaze versus convergence



Original schematic: LEFT INO during RIGHT gaze; patient-side labels are explicit.

Compare the motor signatures

Finding	Clinical inference
INO	Ipsilateral adduction lag + opposite abducting nystagmus. Convergence may be preserved; loss does not exclude INO.
Isolated VI paresis	Weak abduction of one eye; often worse at distance and towards that side.
IV pattern	Hypertropia/excyclotorsion, worse in adduction/down gaze and ipsilateral head tilt.

References AAO: Internuclear ophthalmoplegia | AAO: Fourth nerve palsy

Restriction is not denervation

When the eye cannot elevate

- 1 Look for proptosis, lid retraction and an enlarged inferior rectus in thyroid eye disease.
- 2 A tight inferior rectus prevents passive elevation: specialist forced ductions distinguish mechanical restriction from paresis.
- 3 Investigate the cause before treating the deviation. New neurological signs or optic-nerve dysfunction change urgency.

Avoid a muscle-name shortcut

Problem	Why elevation fails
Tight inferior rectus	Same eye rests low; cannot elevate because the antagonist is tethering it.
Weak elevator	Reduced active elevation; passive movement is usually not mechanically blocked.

Clinical limit

Head-tilt patterns support localisation but are not perfectly specific. Compare ductions, alignment, pupils, lids and the neurological examination before naming an isolated palsy.